

CRC Card Builder Page

CRC Card 1

User

Card Name	User		
Superclasses			
Subclasses	Entrant Organizer Administrator		
Responsibilities	Maintain common user data (name, email, password, etc.) Authenticate via device or Firebase Access personal profile Receive basic system notifications	Collaborators	Profile FirebaseService NotificationManager

CRC Card 2

Entrant

Card Name	Entrant		
Superclasses	User		
Subclasses			
Responsibilities	Join or leave waiting lists Accept/decline invitations View event details Manage personal geolocation permissions Upload or remove profile picture Receive win/lose notifications	Collaborators	Event WaitingList NotificationManager DeviceAuthenticator Profile

CRC Card 3

Organizer

Card Name	Organizer		
Superclasses	User		
Subclasses			
Responsibilities	Create and publish events Generate QR codes for events Upload/update event posters Conduct lottery draws and manage prizes Send notifications to entrants Manage facility profile	Collaborators	Event QRGenerator WaitingList NotificationManager FirebaseService

CRC Card 4

Administrator

Card Name	Administrator		
-----------	---------------	--	--

Superclasses	User		
Subclasses			
Responsibilities	Remove events, profiles, or images Browse events and user data Oversee database consistency Moderate reported content or mis	Collaborators	Event Profile FirebaseService Organizer

CRC Card 5

Event

Card Name	Event		
Superclasses			
Subclasses	TimedEvent GeoVerifiedEvent		
Responsibilities	Store basic event information (title, location, date) Link to waiting list and QR code Maintain registration status (open/closed) Handle cancellations and replacements	Collaborators	Organizer WaitingList LotteryManager FirebaseService QRGenerator

CRC Card 6

GeoVerifiedEvent

Card Name	GeoVerifiedEvent		
Superclasses	Event		
Subclasses			
Responsibilities	Require geolocation verification for entrants Store allowed regions or coordinates	Collaborators	Entrant Organizer WaitingList FirebaseService

CRC Card 7

TimedEvent

Card Name	TimedEvent		
Superclasses	Event		
Subclasses			
Responsibilities	Include recurring schedules (e.g., weekly, monthly) Track session start and end times	Collaborators	Organizer FirebaseService

CRC Card 8

WaitingList

Card Name			
-----------	--	--	--

	WaitingList		
Superclasses			
Subclasses			
Responsibilities	Maintain list of entrants interested in event Handle join/leave operations Manage random selection and re-selection Communicate selection outcome	Collaborators	Entrant Event LotteryManager NotificationManager

CRC Card 9

LotteryManager

Card Name	LotteryManager		
Superclasses			
Subclasses			
Responsibilities	Randomly select entrants from waiting list Manage replacement draws when necessary Notify results via NotificationManager	Collaborators	WaitingList Entrant Event NotificationManager

CRC Card 10

NotificationManager

Card Name	NotificationManager		
Superclasses			
Subclasses			
Responsibilities	Send event updates and lottery results Handle opt-in/out preferences Notify organizers about response	Collaborators	Entrant Organizer Event FirebaseService

CRC Card 11

FirebaseService

Card Name	FirebaseService		
Superclasses			
Subclasses			
Responsibilities	Persist and retrieve all app data (local and remote) Synchronize data in real time Enforce access control by user roles	Collaborators	Event User Profile QRGenerator NotificationManager

CRC Card 12

QEGenerator

Card Name	QEGenerator		
Superclasses			
Subclasses			
Responsibilities	Generate unique QR codes for e Decode scanned codes to acces Store or retrieve QR data from Fi	Collaborators	Event FirebaseService

CRC Card 13

DeviceAuthenticato

Card Name	DeviceAuthenticato		
Superclasses			
Subclasses			
Responsibilities	Identify entrants using unique de Link device ID to profile Enable login without passwords	Collaborators	Entrant FirebaseService

CRC Card 14

Profile

Card Name	Profile		
Superclasses			
Subclasses			
Responsibilities	Manage general profile informati Allow profile updates and image Provide default avatar when miss	Collaborators	User FirebaseService

CRC Card 15

EntrantProfile

Card Name	EntrantProfile		
Superclasses			
Subclasses			
Responsibilities	Store entrant-specific preference Track joined events	Collaborators	Entrant FirebaseService

CRC Card 16

OrganizerProfile

Card Name	OrganizerProfile		
Superclasses			

Subclasses			
Responsibilities	Store facility details (address, de Manage organizer identity and ve	Collaborators	Organizer FirebaseService